Algorithm for Decomposition Using Functional Dependencies DBMS

> **Definition: Decomposition Algorithms** split a complex relational schema R into smaller sub-schemas (R1, R2, ..., Rn) to eliminate anomalies while preserving **Lossless-Join** and **Functional Dependencies**.

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1. Detailed Technical Explanation

1. Attribute Closure Algorithm (F+)

The closure of an attribute set X under F, denoted **X^+**, is the set of all attributes functionally determined by X.

Algorithm Steps:

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1. Set X+ = X

2. Rep

### 2. Lossless-Join Decomposition Testing

A decomposition of R into R1 and R2 is **lossless-join** with respect to F if and only if:

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3. 3NF Synthesis Algorithm (Lossless-Join & Dependency Preserving)

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Input:

1. Compute Minimal Cover Fc for F.

2. For

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## 2. Core Concepts & Memory Keywords

- **Minimal Cover (Canonical Cover):** A simplified set of FDs with single attributes on the right-hand side and no extraneous attributes.

- **L
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3. Must-Write Points for Exams

- Lossless join ensures that joining decomposed tables produces the exact original dataset without fake data.

- Dep

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4. Quick Recall Flow

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Find M
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